

1. **WARM UP: Refresh vocabulary.**
2. **Task 2 – Build a Simple Circuit**

Go to **Tinkercad Circuits** (<https://www.tinkercad.com/circuits>) or **Wokwi Arduino Simulator** (<https://wokwi.com>).

- Create a **new circuit**.
- Drag and drop the components: battery (or power supply), LED, resistor, wires.
- Connect them as in a real circuit.
- Start the simulation and check if the LED lights up.

Circuit 1 – Basic LED Circuit

Components: 1 battery, 2 wires, 1 LED, 1 resistor.

Circuit 1

Place one wire on the negative side of the battery and the other on the 220 Ω resistor.

Place the resistor on the positive side of the LED, and from the negative side of the LED, place one wire that reaches the negative side of the battery.

Circuit 2 – LED with Switch

Components: 1 battery, 3 wires, 1 LED, 1 resistor, 1 switch.

Circuit 2

Connect a wire from the positive end of the battery to the middle pin of the switch. Connect one of the side pins to the resistor, and connect the resistor to the positive lead of the LED. Connect a wire from the negative lead of the LED to the negative end of the battery.

Circuit 3 – Two LEDs in Series

Components: 1 battery, 2 LEDs, 2 resistors, wires.

Circuit 3

Connect a cable from the battery to the first resistor. Then, connect the resistor with an LED from the positive terminal behind another resistor, but connect the resistor to the first LED and the same positive terminal. After the resistor, connect another LED that is connected to

the positive terminal. After the last LED, connect a cable from the negative terminal to the negative terminal of the battery.

Circuit 4 – Two LEDs in Parallel

Components: 1 battery, 2 LEDs, 2 resistors, wires.

Circuit 4

Connect a wire from the positive side of the battery to the resistor. Then, connect the resistor to the positive side of the LED. Connect a wire from the negative side of the LED to the negative side of the battery. Repeat the procedure again.

3. Task 3 – Describe Your Actions (Imperatives)

- Write the instructions in English while you build

4. Task 4 – Describe How It Works (Present Simple)

- Write **3 sentences** about your circuit using Present Simple:

E.g: The battery provides power.

5. MATCH THE WORD WITH THE DEFINITION

- Battery c
- LED e
- Wire a
- Circuit f
- Conductor b
- Insulator d

Set B – Definitions (to match)

- a_ A thin metal strand that carries electricity
- b_ A material that allows electricity to pass (e.g., copper).
- c_ Stores and provides electricity.
- d_ A material that does not allow electricity to pass (e.g., plastic)
- e_ Small light that uses little energy.
- f_ A closed path for electricity to flow.

6. Write TRUE or FALSE. Correct the false.

Electricity flows in an open circuit.**false**

In an open circuit, no electricity flows because the path is interrupted.

Plastic is an insulator.**true**

A battery is a conductor.**false**

A battery is a power source (generator), not a conductor.

A closed circuit allows electricity to flow.**true**